Artificial Intelligence Lab

Lab Report # 01

Submitted to:

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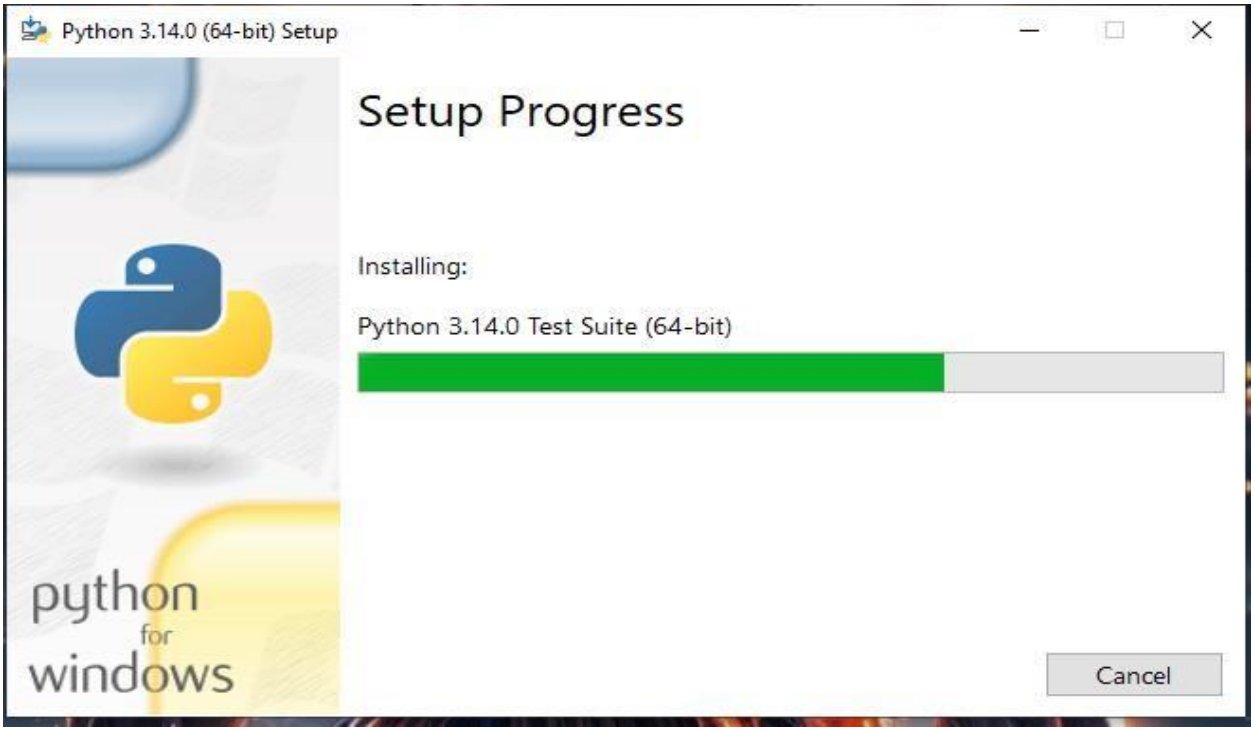
Objectives:

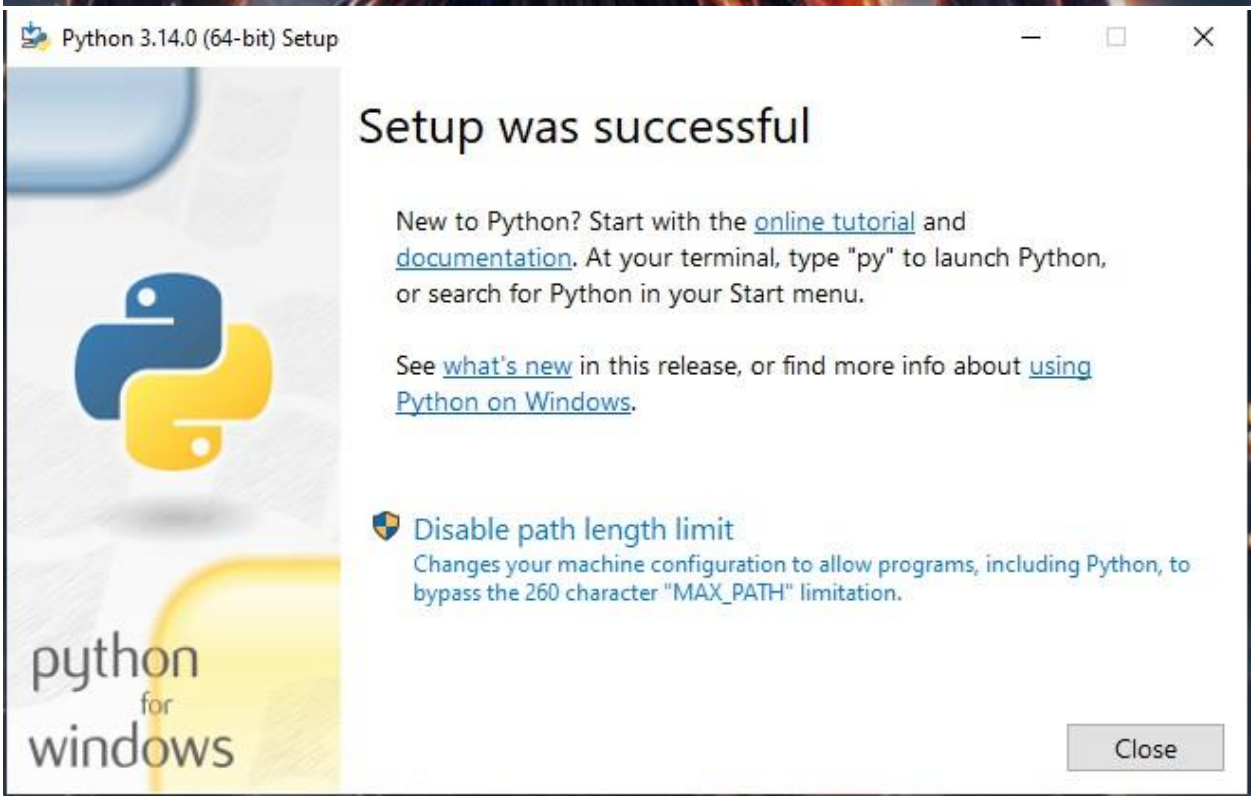
1. Setting up the development environment.
2. Getting started with Python
3. Understanding basics of Python programming language.

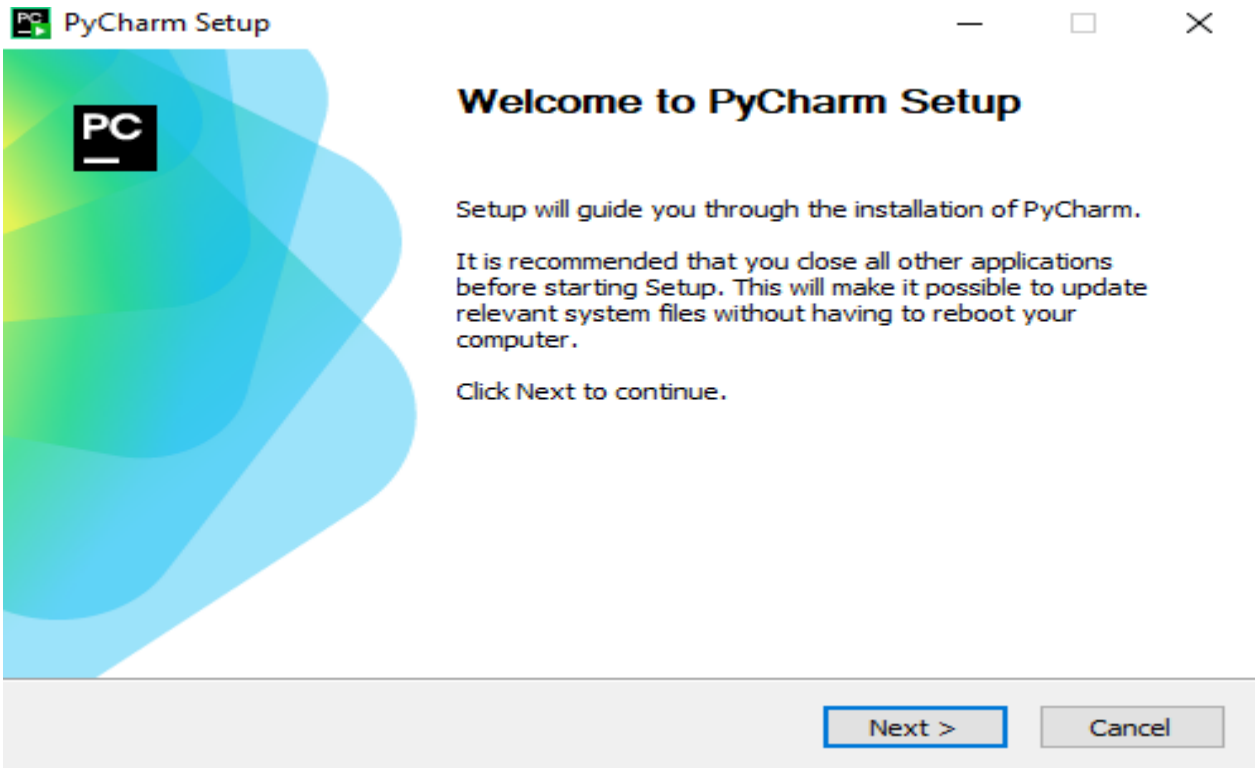
Lab Task 1

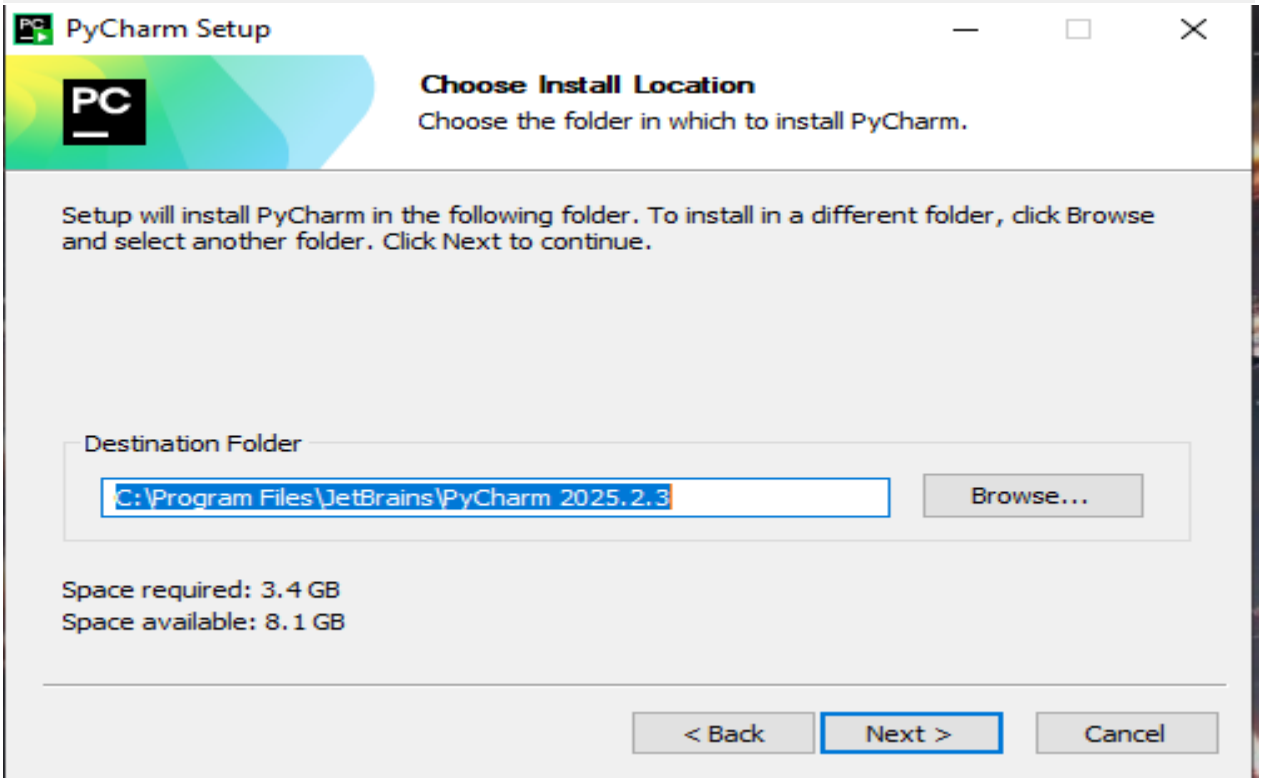
Introduction to Python and Installation

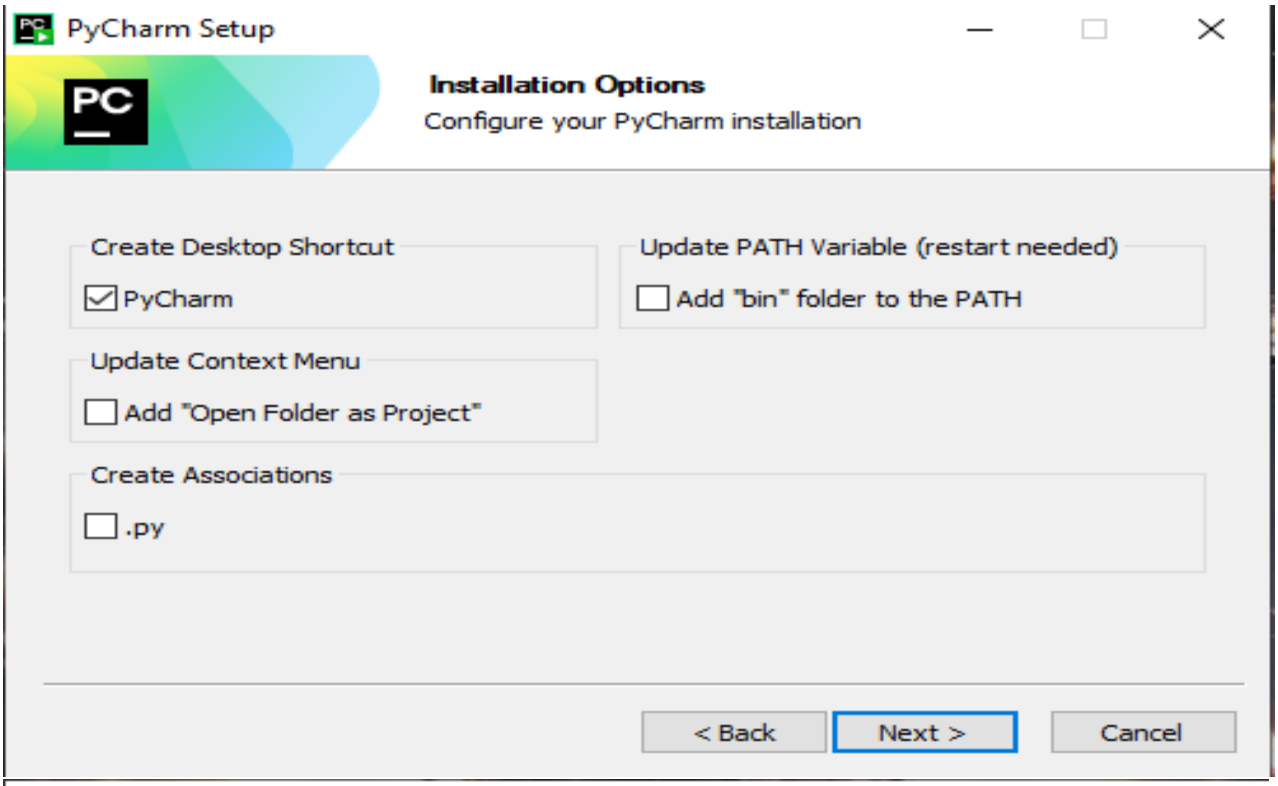
Solution: -

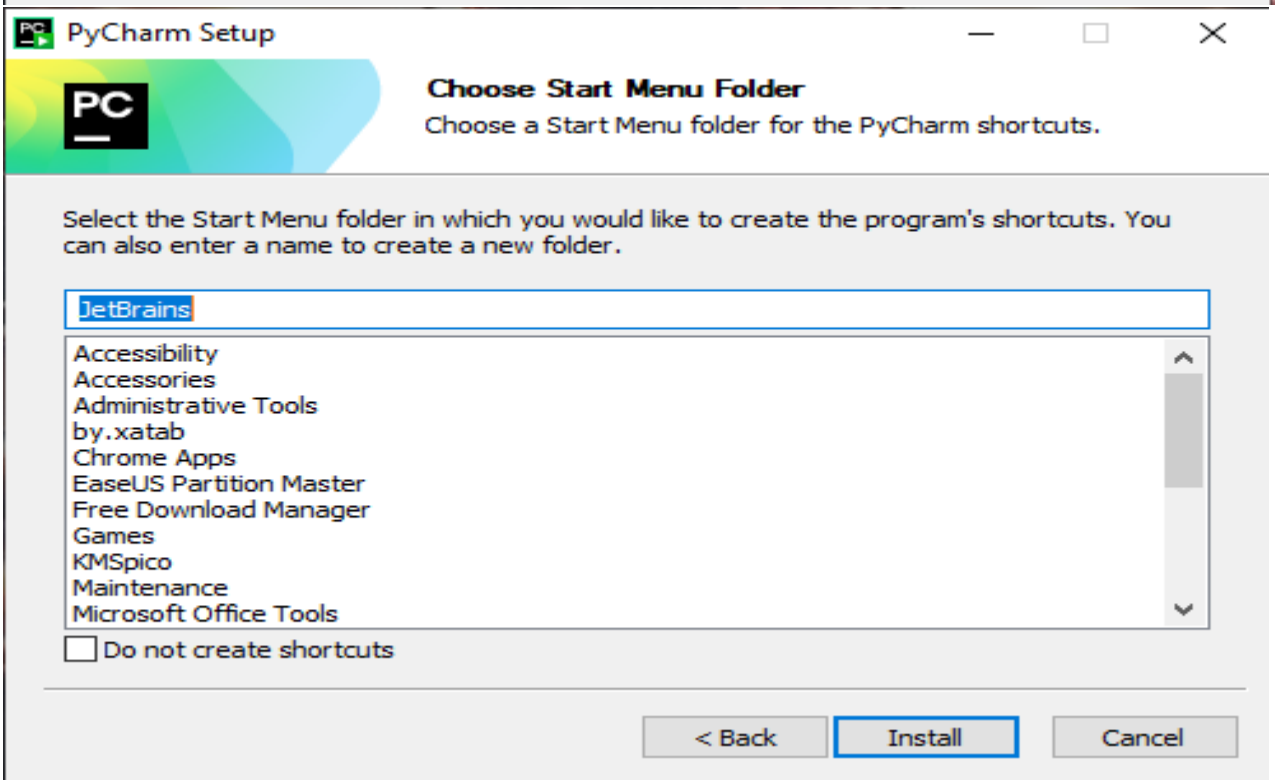

Python 3.14.0 (64-bit) Setup
Setup Progress
Installing:
Python 3.14.0 Test Suite (64-bit)
python
for
windows
Cancel

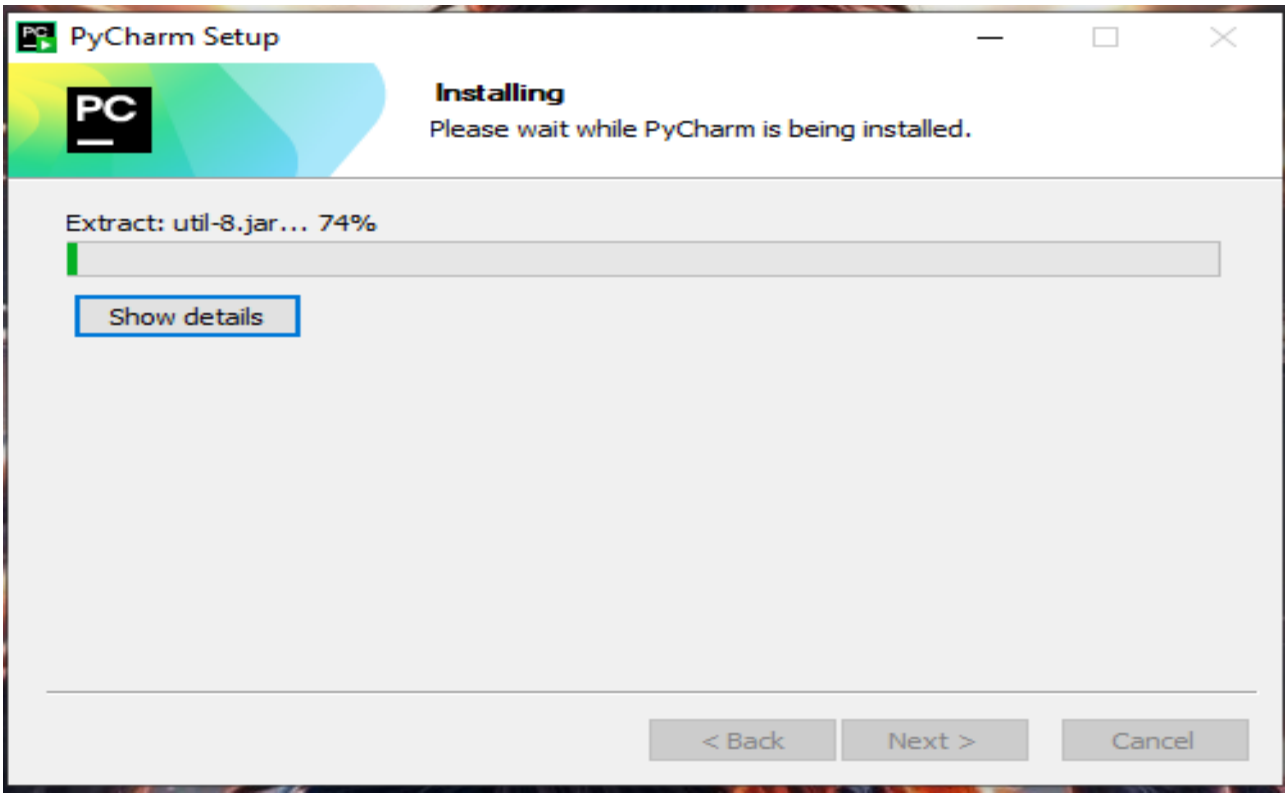

Python 3.14.0 (64-bit) Setup
Setup was successful
New to Python? Start with the online tutorial and documentation. At your terminal, type "py" to launch Python, or search for Python in your Start menu.
See what's new in this release, or find more info about using Python on Windows.
Disable path length limit
Changes your machine configuration to allow programs, including Python, to bypass the 260 character "MAX_PATH" limitation.
python
for
windows
Close

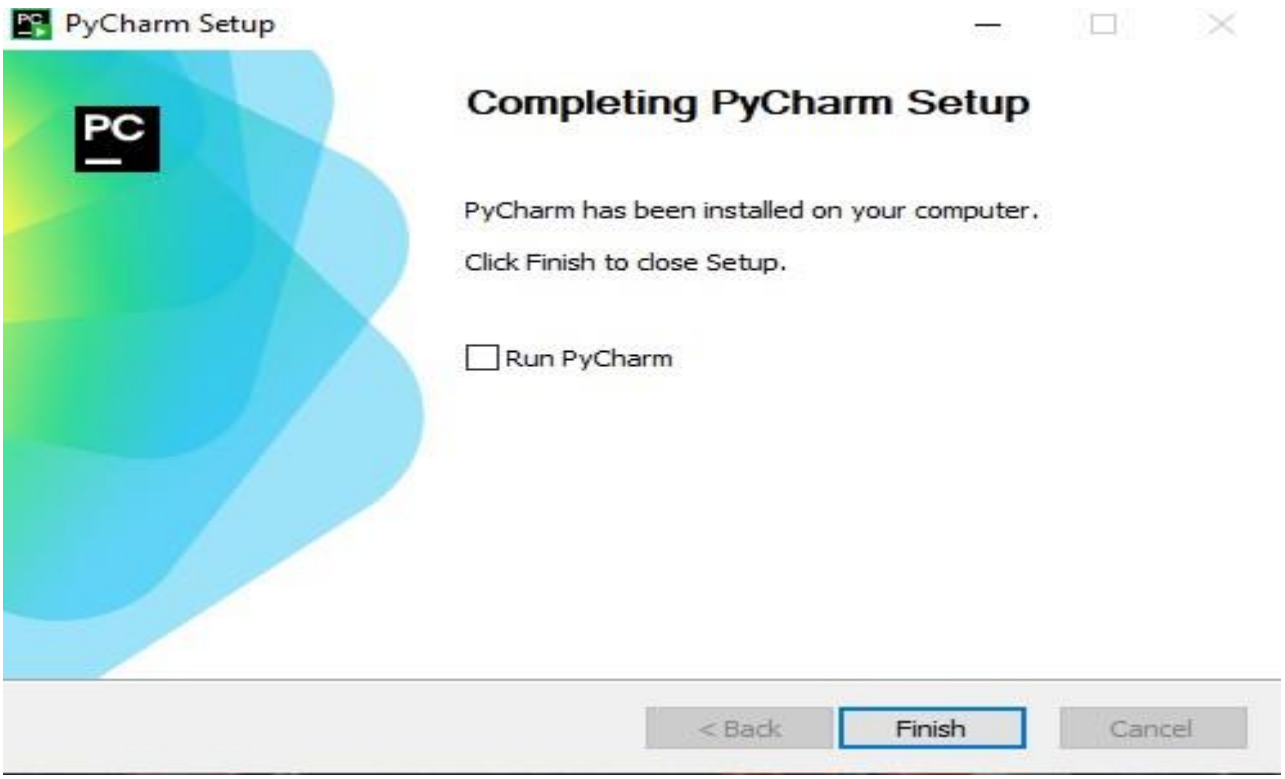

PyCharm Setup
Welcome to PyCharm Setup
Setup will guide you through the installation of PyCharm.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel


PyCharm Setup
Choose Install Location
Choose the folder in which to install PyCharm.
Setup will install PyCharm in the following folder. To install in a different folder, click Browse and select another folder. Click Next to continue.
Destination Folder
C:\Program Files\JetBrains\PyCharm 2025.2.3
Browse...
Space required: 3.4 GB
Space available: 8.1 GB
< Back
Next >
Cancel


PyCharm Setup
Installation Options
Configure your PyCharm installation
Create Desktop Shortcut
PyCharm
Update PATH Variable (restart needed)
Add "bin" folder to the PATH
Update Context Menu
Add "Open Folder as Project"
Create Associations
.py
< Back
Next >
Cancel


PyCharm Setup
Choose Start Menu Folder
Choose a Start Menu folder for the PyCharm shortcuts.
Select the Start Menu folder in which you would like to create the program's shortcuts. You can also enter a name to create a new folder.
JetBrains
Accessibility
Accessories
Administrative Tools
by.xatab
Chrome Apps
EaseUS Partition Master
Free Download Manager
Games
KMSpico
Maintenance
Microsoft Office Tools
Do not create shortcuts
< Back
Install
Cancel


PyCharm Setup
Installing
Please wait while PyCharm is being installed.
Extract: util-8.jar... 74%
Show details
< Back
Next >
Cancel


PyCharm Setup
Completing PyCharm Setup
PyCharm has been installed on your computer.
Click Finish to close Setup.
Run PyCharm
< Back
Finish
Cancel

Lab Task # 02

Question # 01

Write a Python Code to write the table of a number given by the user.

Solution: -

```python
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Output: -

```
Enter a number:   8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80

Process finished with exit code 0
```

Question # 02

Write a python code to check wheatear a number is prime or not.

Solution: -

```python
num = int(input("Enter a number: "))

if num > 1:
    for i in range(2, num):
        if num % i == 0:
            print(num, "is not a prime number.")
            break
    else:
        print(num, "is a prime number.")
else:
    print(num, "is not a prime number.")
```

Output: -

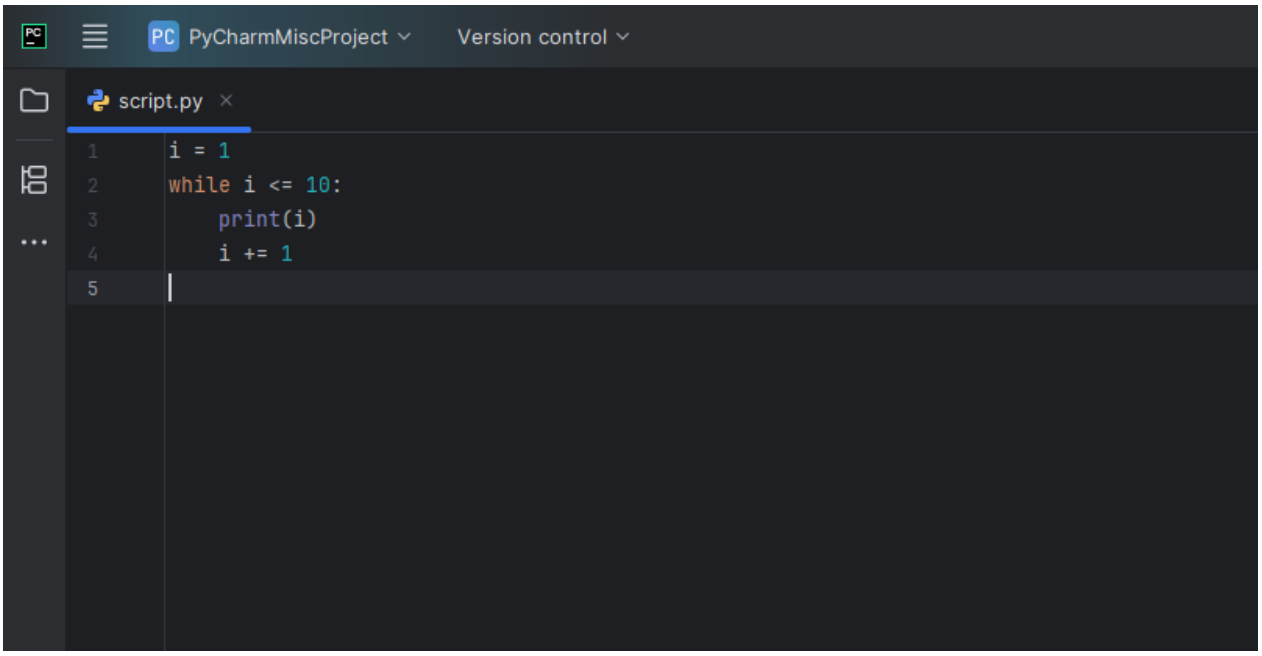
```
Enter a number: 5
5 is a prime number.

Process finished with exit code 0
```

Question # 03a

Write a Python program that prints numbers from 1 to 10 using a while loop. Modify the while loop to print only even numbers between 1 and 20.

Solution: -



Output: -

Question # 03b

Write a Python program that prints numbers from 1 to 10 using a while loop. Modify the while loop to print only even numbers between 1 and 20.

Solution: -

```python
i = 1
while i <= 20:
    print(i)
    i += 1
```

Output: -

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Process finished with exit code 0
```

Question # 04

Write a Python program to determine whether the given number is a Harshad Number. If a number is divisible by the sum of its digits, then it will be known as a Harshad Number. For example: The number 156 is divisible by the sum (12) of its digits (1, 5, 6) Some Harshad numbers are 54, 120, etc.

Solution: -

```python
num = int(input("Enter a number: "))
sum_of_digits = sum(int(digit) for digit in str(num))

if num % sum_of_digits == 0:
    print(num, "is a Harshad Number.")
else:
    print(num, "is not a Harshad Number.")
```

Output: -

```
Enter a number: 45
45 is a Harshad Number.

Process finished with exit code 0
```